

ANKIT SINGH

Full-Stack Platform Engineer | API & System Performance | AWS | Reliability, Observability & Incident Response

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Lead engineer with 6+ years of experience building and scaling customer-facing systems and APIs in fintech and cloud environments. Strong in system design, data modeling, reliability/observability, and performance optimization. Experienced in end-to-end architecture and production operations (monitoring, incident response, cost and performance tuning) and working with product/ops to ship useful features. I enjoy contributing to open source projects in my free time.

Senior Software Engineer — Sequoia July 2024 - Present
Project: GenAI platform (Dify + RAG)

- **Built** and maintained a **reusable internal GenAI agent platform (Dify + RAG)**, with shared configs, runbooks, and guardrails, **using Python, FastAPI and MongoDB**. Focused on the relevance of retrieved context and quality of responses for internal users.
- Refined **prompts and retrieval in Dify's Prompt IDE** to make answers more relevant and reduce bad responses.
- Led **technical decision-making and architectural tradeoffs**, aligned engineering work with business outcomes and system constraints.

Lead Engineer — LotusPay (acquired by Juspay) February 2020 - June 2024
Project: Developing India's leading NACH mandate and payments system

Public Platform & Developer Experience

- Owned **payment dashboards** using **Ruby on Rails and JavaScript** for **CRUD payment operations**, **simplifying workflows** for **support and ops teams**.
- Developed **targeted payment tracking pop-ups**, improving user clarity and **increasing payment completion from 69% to 90%** through enhanced visibility into transaction statuses.
- Optimized **AJAX requests** through **HTTP/2**, **enabled keep-alive**, and **batched requests**, helping **reduce page load time from 7s to 1s**, delivering a smoother user experience.

Reliability, Observability & Incident Response

- Engineered **scalable REST APIs (Rails, Redis, AWS EC2)**, improving **payment success rate from 50% to 90%**; implemented consistent error schemas, **pagination**, and **API versioning to reduce integration issues** and support escalations.
- Improved **API developer** experience by standardizing responses and logs, removing customer confusion and lifting CSAT by **25%**.
- Fixed critical endpoints with **unreliable client/network conditions** with **idempotency keys** and **strict request validation** to **prevent duplicate processing** and inconsistent states under retries.
- Introduced **retry-safe processing**, **traceable logging**, and **audit-friendly** safeguards in core payment flows, plus data-driven changes that **increased conversions by 30%**.
- Improved **performance with caching**, tuned **DB queries/indexes**, and **blue-green deployments**, and set up CloudWatch + SNS + Lambda alerts to **cut downtime by 95%** and keep incident resolution under **5 minutes**.
- Reworked a high-throughput bank **payment data ingestion pipeline** by running **A/B tests on batch size, indexing, and concurrency settings** on **AWS RDS (MySQL)**, the winning setup cut **processing time by 15x** and made imports far more reliable under load.

Performance & Release Engineering

- Redesigned the **core MySQL schema on AWS RDS**, tuned **heavy queries (New Relic + EXPLAIN)**, and added **Redis + CDN caching**, cutting server load by **50%** and saving over **\$10k annually**.
- Implemented **blue-green deployments** using **Route 53 + S3 + CloudFront**, enabling **zero-downtime rollouts** and reducing release-related disruptions by **30% during high-traffic hours**.
- **Led quarterly reliability and performance efforts** and **mentored engineers** through design and code reviews, improving code quality and release practices.

Skills

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| ● Ruby on Rails, Python, Node.js | ● Flask, FastAPI, Shell Scripting (Linux/Ubuntu), REST APIs |
| ● MySQL, PostgreSQL, MongoDB | ● Redis, Sidekiq, Git, GitHub, OpenAI APIs |
| ● HTML, CSS, React, WebSocket (Real-time info) | ● Certified Amazon Web Services Solution Architect |
| ● Grafana, Prometheus, AppSignal, New Relic | ● RSpec, PyTest, Microservices, Monorepo |
| ● Learning: TypeScript, GraphQL, Jest, Kubernetes | ● Docker |

Achievements

- Open Source contributor to [Polar.sh](#)
- Published manuscript on [Object Detection and Segmentation](#) in an International Scientific Research Journal.
- Successfully got the first startup acquired by Juspay

Education

Bachelor of Technology in Computer Science at Cambridge Institute of Technology (2016-2020)